

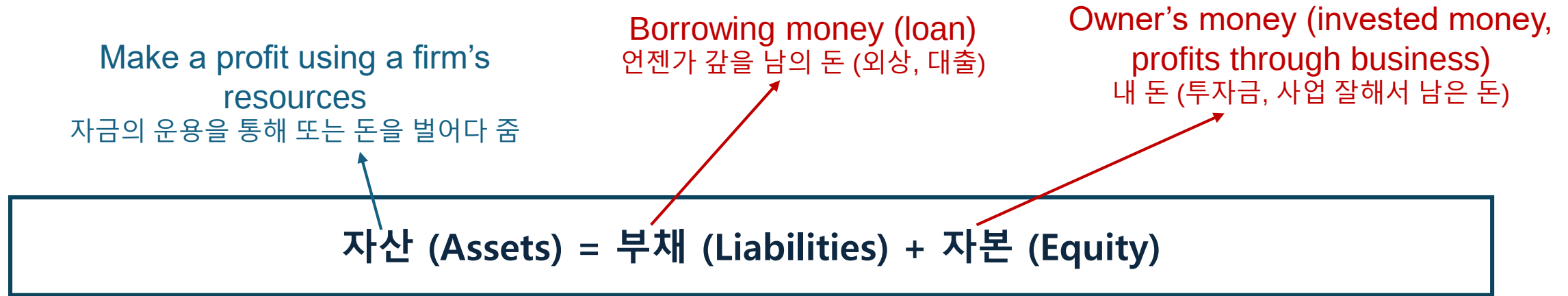
Accounting Equation

Principles of Accounting
Dr. Joe (Joong Hi) Cho

Accounting Equation and Components 2 (Understanding of 'Equity')

회계 등식과 구성요소 2
(‘자본’ 구성요소에 대한 이해)

Accounting Equation



- Assets: Resources owned or controlled by the company 기업이 보유한 자원
- Liabilities: Debts and obligations the company must repay 기업이 갚아야 할 빚과 의무(빌린 남의 돈)
- Equity: Owner's residual interest in the company (Equity = Assets – Liabilities)
총자산에서 빌린 돈을 제외한 주인이 실제로 소유한 순수한 몫 (=자산 – 부채)

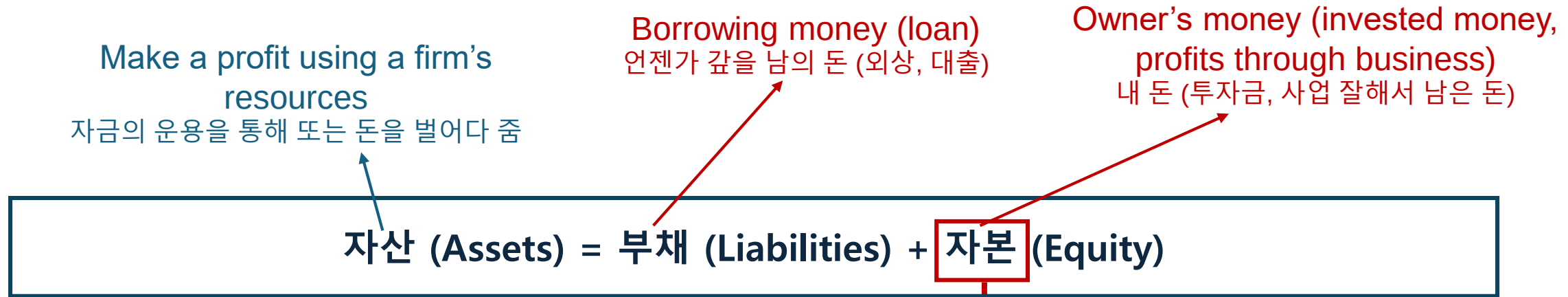
Q) If Apple sells iPhones and earns profit, where does the money go?

만약, 애플이 아이폰을 팔아서 수익이 생긴다면 그 돈은 어디로 갈까?

A) Since the revenue belongs to the owner, it is recorded as Equity

그 수익은 내 돈(=주인 돈) 이기 때문에 '자본'으로 간다.

Components of Equity



- Equity consists of two main components 자본은 크게 두 갈래로 나눠 설명된다.
 - Money invested by the owner (shareholders) → **Contributed Capital**
주인이 직접 넣은 돈 → 자본금(Contributed Capital)
ex) When starting the company, the owner invests \$1,000
(i.e., issuing \$1,000 of shares)
회사를 시작할 때 주인이 100만원 넣음 (다시 말해, 주식 100만원 발행)
 - Profits earned from business operations → **Retained Earnings**
사업을 해서 벌어들인 돈 → 이익잉여금(Retained Earnings)
ex) Profit from selling iPhones remains in the company
아이폰을 팔아서 남은 돈이 회사 안에 쌓임

Components of Equity

Owner's money (invested money, profits through business)
내 돈 (투자금, 사업 잘해서 남은 돈)

$$\text{자산 (Assets)} = \text{부채 (Liabilities)} + \text{자본 (Equity)}$$

- Equity consists of two main components 자본은 크게 두 갈래로 나눠 설명된다.
 1. Money invested by the owner (shareholders) → Contributed Capital
ex) When starting the company, the owner invests \$1,000 (i.e., issuing \$1,000 of shares)
 2. Profits earned from business operations → Retained Earnings
ex) Profit from selling iPhones remains in the company

- A business generates profit when earnings exceed costs
사업을 해서 이익을 남긴다는 것은 장사해서 '번 돈'이 '쓴 돈' 보다 많아야 함
- In accounting, profit occurs when Revenue > Expenses
회계적으로, 수익(Revenue)이 비용(Expense)보다 많은 경우를 뜻함
- ex) \$1,000(Sales Rev) – \$400(Material Exp) – \$200(Rent Exp)
= \$400(Net Income)
매출 100만원(수익) – 재료비 40만원(비용) – 임대료 20만원(비용) = 40만원 순이익(Net Income)

Accounting Equation

- Therefore, a new accounting equation is as follows:

Make a profit using a firm's
resources
자금의 운용을 통해 또는 돈을 벌어들임

Borrowing money (loan)
언젠가 갚을 남의 돈 (외상, 대출)

Owner's money (invested money,
profits through business)
내 돈 (투자금, 사업 잘해서 남은 돈)

$$\text{자산 (Assets)} = \text{부채 (Liabilities)} + \text{자본 (Equity)}$$

Accounting Equation

- Therefore, we can re-write accounting equation as follows:

Make a profit using a firm's
resources
자금의 운용을 통해 돈을 벌어들임

Borrowing money (loan)
언젠가 갚을 남의 돈 (외상, 대출)

Owner's money (invested money,
profits through business)
내 돈 (투자금, 사업 잘해서 남은 돈)

$$\text{자산 (Assets)} = \text{부채 (Liabilities)} + \text{자본 (Equity)}$$

Owner investment

Profits from business operations

$$\text{자산(Assets)} = \text{부채(Liabilities)} + \text{납입자본(Contributed Capital)} + \text{이익잉여금(Retained Earnings)}$$

Accounting Equation

- Retained Earnings = Revenue - Expenses

자금의 운용(uses)을 통해
돈을 벌어서
Make a profit using a firm's
resources

언젠가 갚을 남의 돈
(외상, 대출)

Liabilities: Money owed to others
(credit, loans)

내 돈

(투자금, 사업 잘해서 남은 돈)
Equity: Owners' money (invested
money, profits through business)

$$\text{자산 (Assets)} = \text{부채 (Liabilities)} + \text{자본 (Equity)}$$

Owner's money
(Invested money)

Owner's money
(profits through business)

$$\text{자산(Assets)} = \text{부채(Liabilities)} + \text{납입자본(Contributed Capital)} + \text{이익잉여금(Retained Earnings)}$$

The money left from business operations = Revenue - Expenses

Accounting Equation

- So far, we can write the accounting equation as follows:

$$\text{자산(Assets)} = \text{부채(Liabilities)} + \text{납입자본(Contributed Capital)} + \text{이익잉여금(Retained Earnings)}$$

$$\text{자산(Assets)} = \text{부채(Liabilities)} + \text{납입자본(Contributed Capital)} + \text{수익(Revenue)} - \text{비용(Expense)}$$

- Shareholders, as the owners of the company, can receive money when the company earns profits

dividends

Shareholders(=investors)
generally have two different
things

→ Stocks (or Shares) = money invested by shareholders

→ Dividends = profits returned to shareholders

Accounting Equation

자산(Assets) = 부채(Liabilities) + 납입자본(Contributed Capital) + 이익잉여금(Retained Earnings)

자산(Assets) = 부채(Liabilities) + 납입자본(Contributed Capital) + 수익(Revenue) - 비용(Expense)

- Consequently, dividends reduce equity because profits from running a business are distributed to shareholders

자산(Assets) = 부채(Liabilities) + 납입자본(Contributed Capital) + 수익(Revenue) - 비용(Expense) - 배당금(Dividends)

Accounting Equation

$$\text{자산 (Assets)} = \text{부채 (Liabilities)} + \text{자본 (Equity)}$$

$$\text{자산(Assets)} = \text{부채(Liabilities)} + \text{납입자본(Contributed Capital)} + \text{수익(Revenue)} - \text{비용(Expense)} - \text{배당금(Dividends)}$$

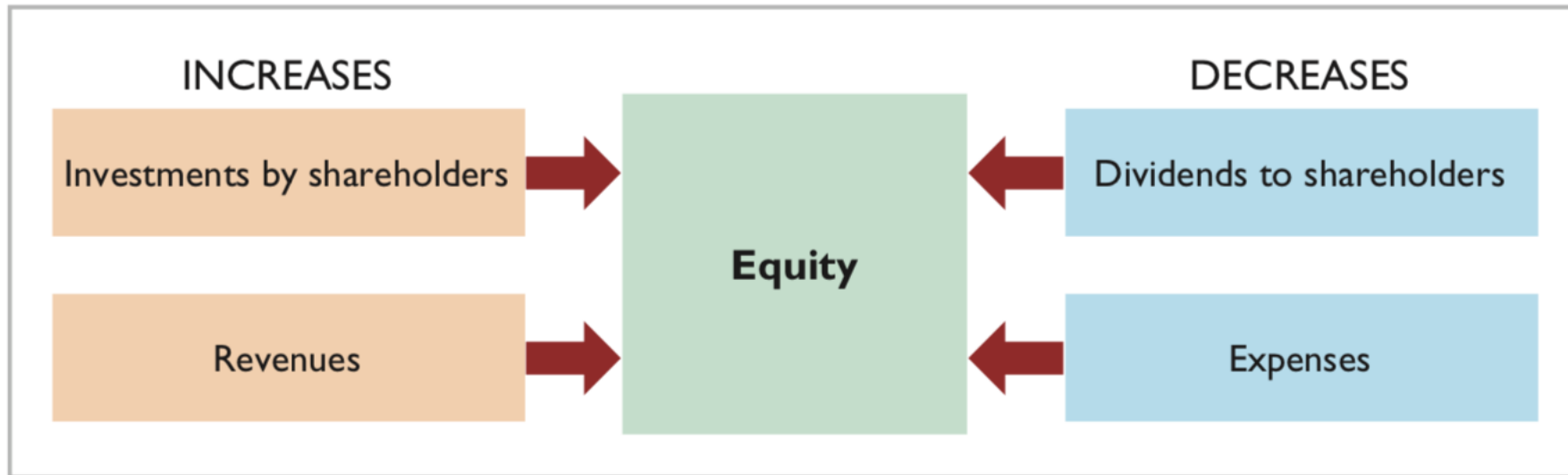


Illustration 1.7: Increases and decreases in equity.

Equity

- **Share capital—ordinary:** describes the amounts paid in by shareholders for the ordinary shares they purchase.
- **Revenues:** are the gross increases in equity resulting from business activities entered into for the purpose of earning income. Revenues usually result in an increase in an asset.
- **Expenses:** are the cost of assets consumed or services used in the process of earning revenue.
- **Dividends:** are distribution of cash or other assets to shareholders. They are not expenses.

Let's see how can we apply the rule of debit and credit based on the details of accounting equation

차변/대변 공식을 구체적인 회계 등식을 바탕으로
다시 이해해보자

Rules for Debit and Credit Recording

- The concepts of double-entry bookkeeping and duality of transactions ultimately lead to the rules of debit and credit recording
- However, the direction of debit and credit entries differs depending on the nature of the account



Q) Why is an increase in assets recorded as debit, and a decrease as credit?

A) That is just a rule. We keep following this rule for make the accounting equation balanced

Rules for Debit and Credit Recording

자산 (Assets)		=	부채 (Liabilities)		+	자본 (Equity)	
↑	↓		↓	↑		↓	↑
Debit	Credit		Debit	Credit		Debit	Credit
+	-		-	+		-	+

자산 (Assets)		=	부채 (Liabilities)		+	주식자본 (Share Capital)		+	수익 (Revenue)		-	비용 (Expense)		-	배당금 (Dividends)	
↑	↓		↓	↑		↓	↑		↓	↑		↑	↓		↑	↓
Debit	Credit		Debit	Credit		Debit	Credit		Debit	Credit		Debit	Credit		Debit	Credit
+	-		-	+		-	+		-	+		+	-		+	-

It is important to understand all types of financial statements; particularly about Retained Earnings

완벽한 차변/대변 공식을 이해하기 위해서는
'재무제표'들을 잘 이해해야 한다. 특히 Retained
Earnings에 대해서

Financial Statements

- The below is the accounting equation that we have learned so far
- More importantly, we must know all types of financial statements, and their interrelationships

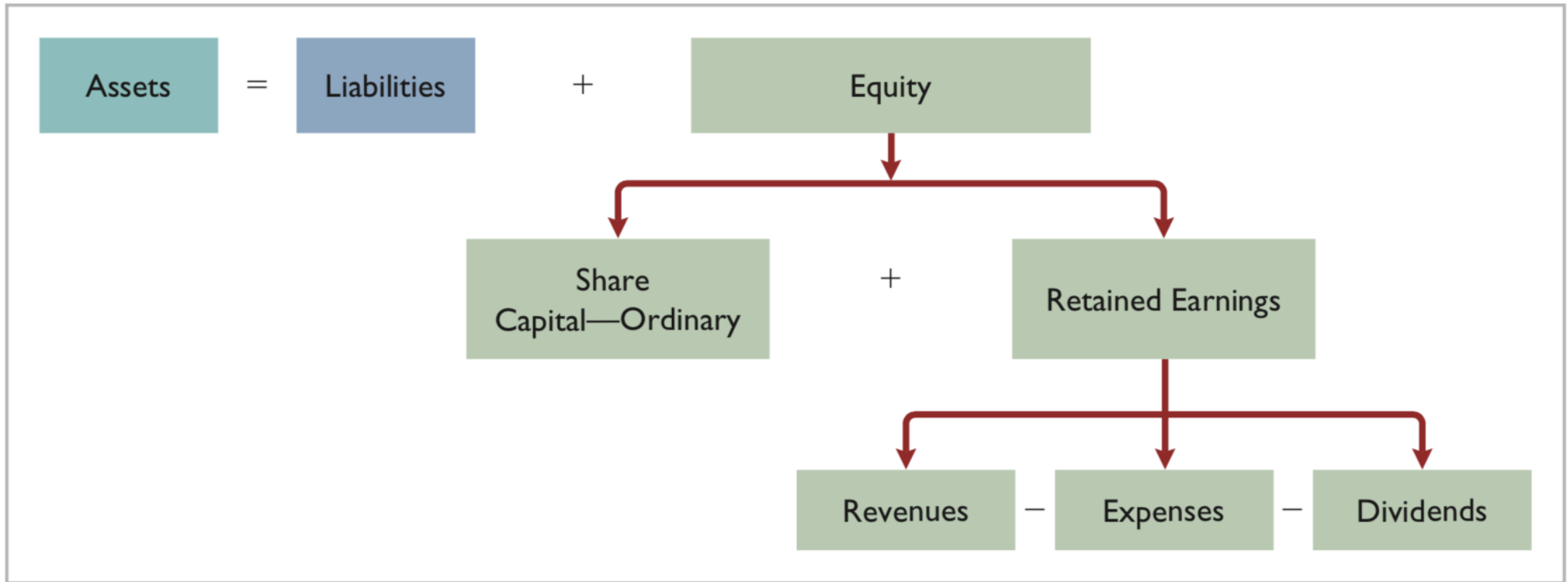


Illustration 1.9: Expanded accounting equation.

Financial Statements

1. **Income statement:** presents the revenues and expenses and resulting net income or net loss for a specific **period of time**.
2. **Retained earnings statement:** summarizes the changes in retained earnings for a specific **period of time**.
3. **Statement of financial position:** reports the assets, liabilities, and equity of a company at **a specific date**. (Sometimes referred to as a *balance sheet*.)
4. **Statement of cash flows:** summarizes information about the cash inflows (receipts) and outflows (payments) for a specific **period of time**.

Financial Statements

Income Statement



Retained Earnings Statement



Statement of Financial Position (Balance Sheet)



Statement of Cash Flows

Softbyte SA Income Statement For the Month Ended September 30, 2020		
Revenues		
Service revenue		€4,700
Expenses		
Salaries and wages expense	€900	
Rent expense	600	
Advertising expense	250	
Utilities expense	200	
Total expenses		<u>1,950</u>
Net income		<u>€2,750</u>
Softbyte SA Retained Earnings Statement For the Month Ended September 30, 2020		
Retained earnings, September 1	€ 0	
Add: Net income	<u>2,750</u>	
	2,750	
Less: Dividends	<u>1,300</u>	
Retained earnings, September 30		<u>€1,450</u>
Softbyte SA Statement of Financial Position September 30, 2020		
<u>Assets</u>		
Equipment	€ 7,000	
Supplies	1,600	
Accounts receivable	1,400	
Cash	<u>8,050</u>	
Total assets		<u>€18,050</u>
<u>Equity and Liabilities</u>		
Equity		
Share capital—ordinary	€ 15,000	
Retained earnings	<u>1,450</u>	
		€16,450
Liabilities		
Accounts payable		<u>1,600</u>
Total equity and liabilities		<u>€18,050</u>
Softbyte SA Statement of Cash Flows For the Month Ended September 30, 2020		
Cash flows from operating activities		
Cash receipts from revenues	€ 3,300	
Cash payments for expenses	<u>(1,950)</u>	
Net cash provided by operating activities		1,350
Cash flows from investing activities		
Purchase of equipment		<u>(7,000)</u>
Cash flows from financing activities		
Sale of ordinary shares	€15,000	
Payment of cash dividends	<u>(1,300)</u>	
		13,700
Net increase in cash		8,050
Cash at the beginning of the period		0
Cash at the end of the period		<u>€ 8,050</u>

Net income is computed first and is needed to determine the ending balance in retained earnings.

The **ending balance in retained earnings** is needed in preparing the statement of financial position.

The **cash** shown on the statement of financial position is needed in preparing the statement of cash flows.

Illustration 1.11: Financial statements and their interrelationships

Financial Statements – Equity Relationships

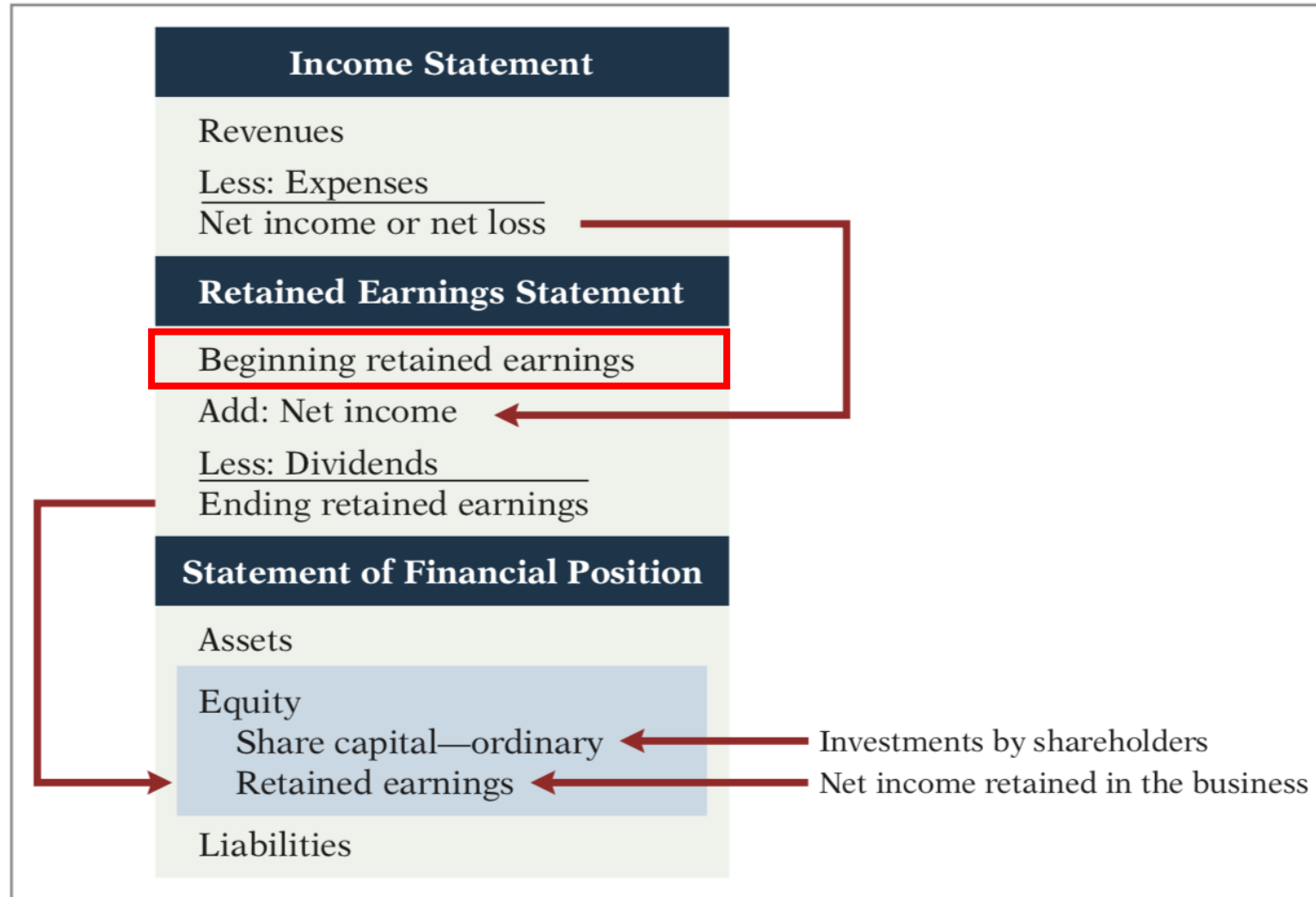
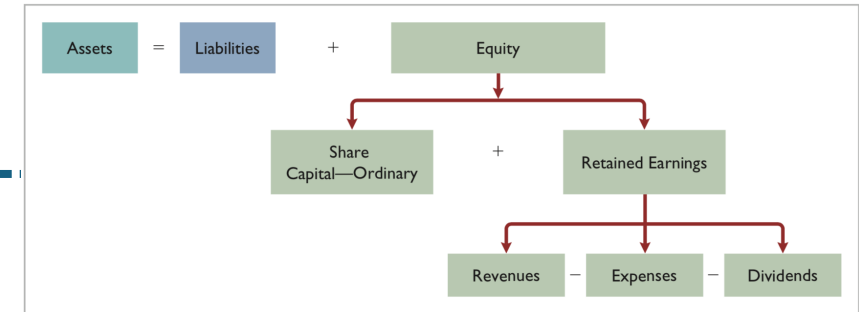


Illustration 2.11: Equity relationships

Financial Statements – Accounting Equation

- The Account (계정)
 - Assets (Cash, ...) / Liabilities (Note Payable, ...) / Shareholder's Equity (Share Capital, ...)
 - Debit and Credit Columns (차변과 대변)
 - Double-Entry System (복식부기)



Assets = Liabilities + Equity

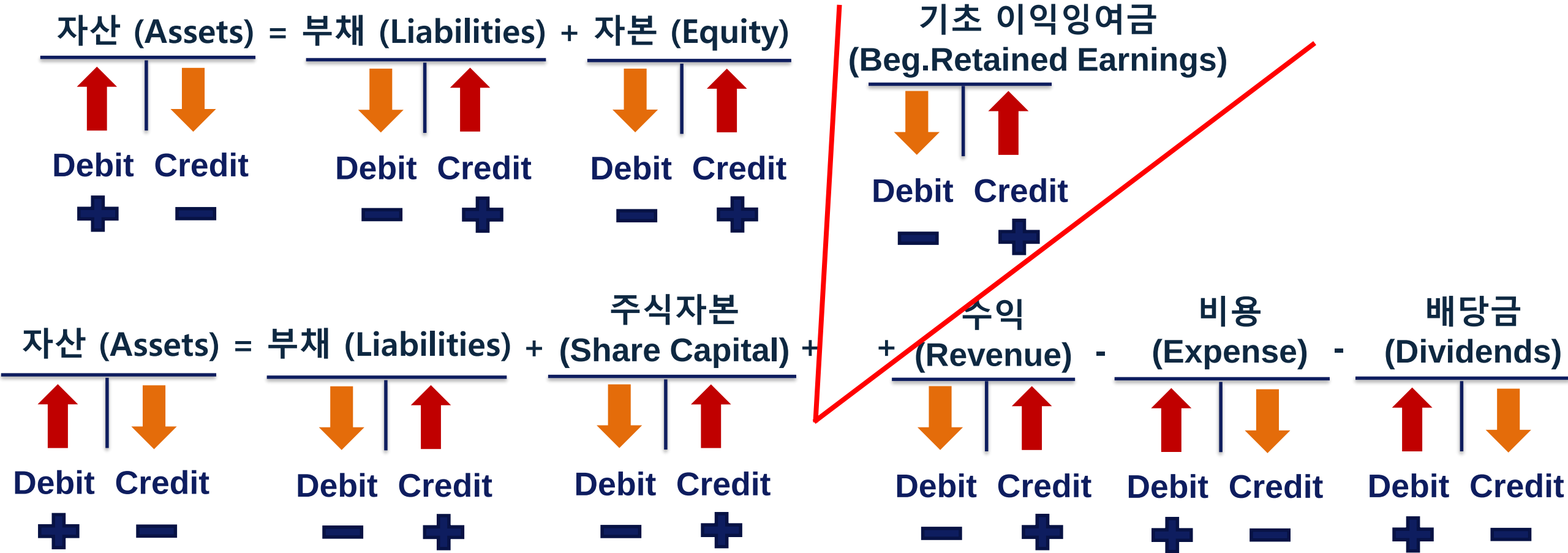
Assets = Liabilities + Share Capital + R/E

Assets = Liabilities + Share Capital + Beg.R/E + Net Income – Dividends

Assets = Liabilities + Share Capital + Beg.R/E + Revenues – Expenses – Dividends

Assets + Expenses + Dividends = Liabilities + Share Capital + Beg.R/E + Revenues

Rules for Debit and Credit Recording



Rules for Debit and Credit Recording

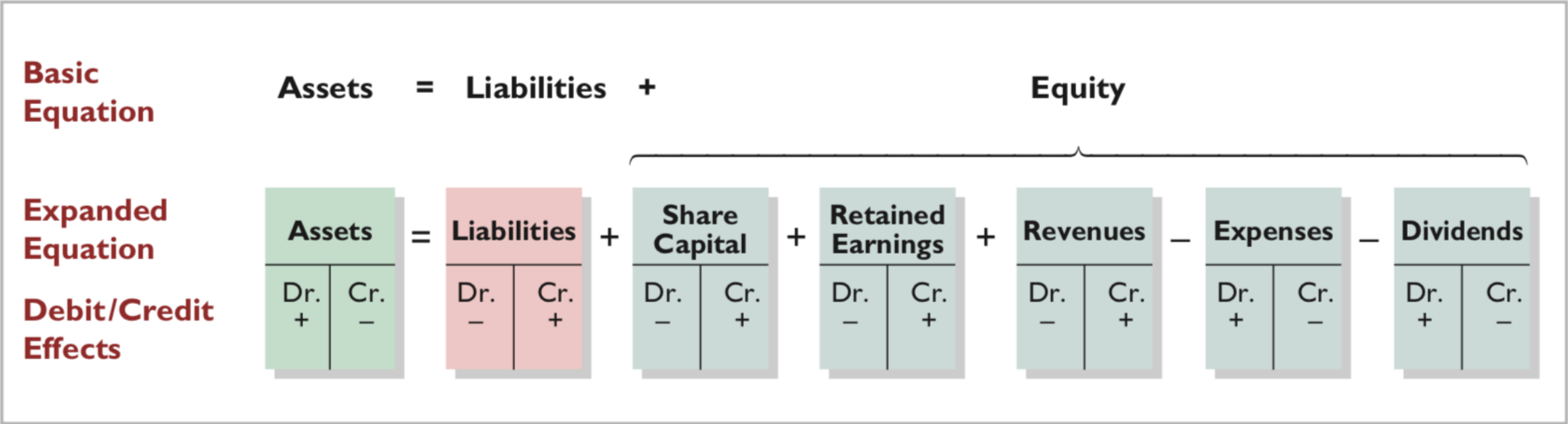


Illustration 2.12: Summary of debit/credit rules